Ankle Joint Assessment

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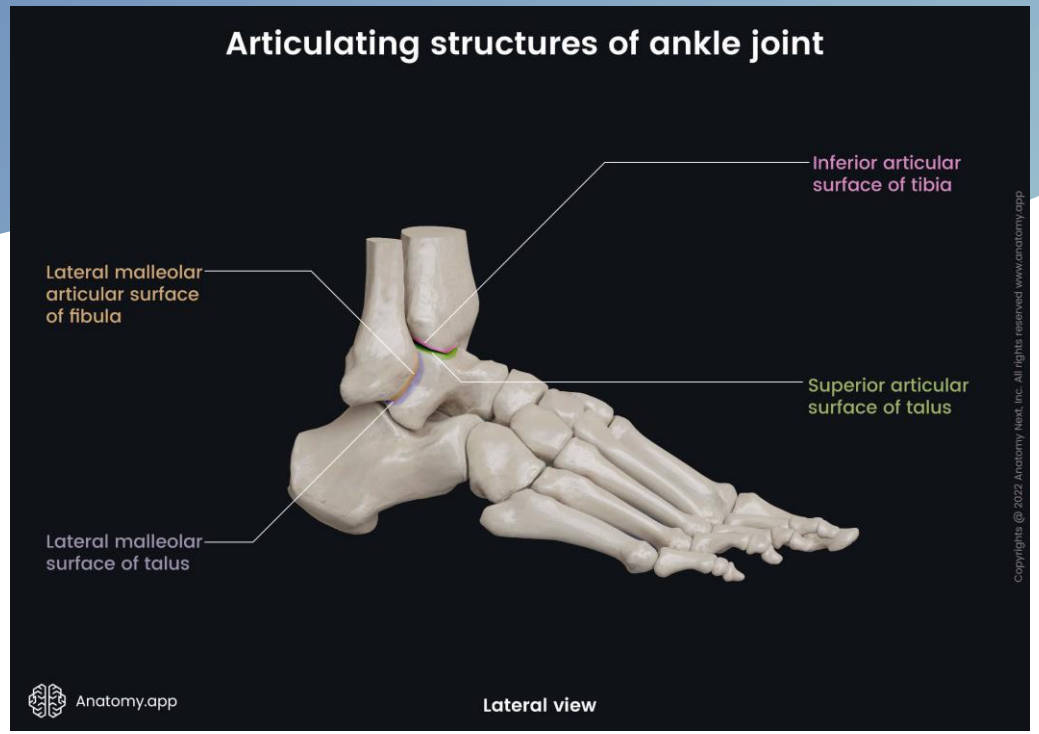
Learning Outcomes

By the End of this lesson, you should be able to:

- Practice the objective MSK assessment of the Ankle joint.



Rearfoot	Midfoot	Forefoot
Talocrural joint	Talonavicular joint	Tarsometatarsal joints
Subtalar joint	Calcaneocuboid joint	Metatarsophalangeal joint Interphalangeal joints

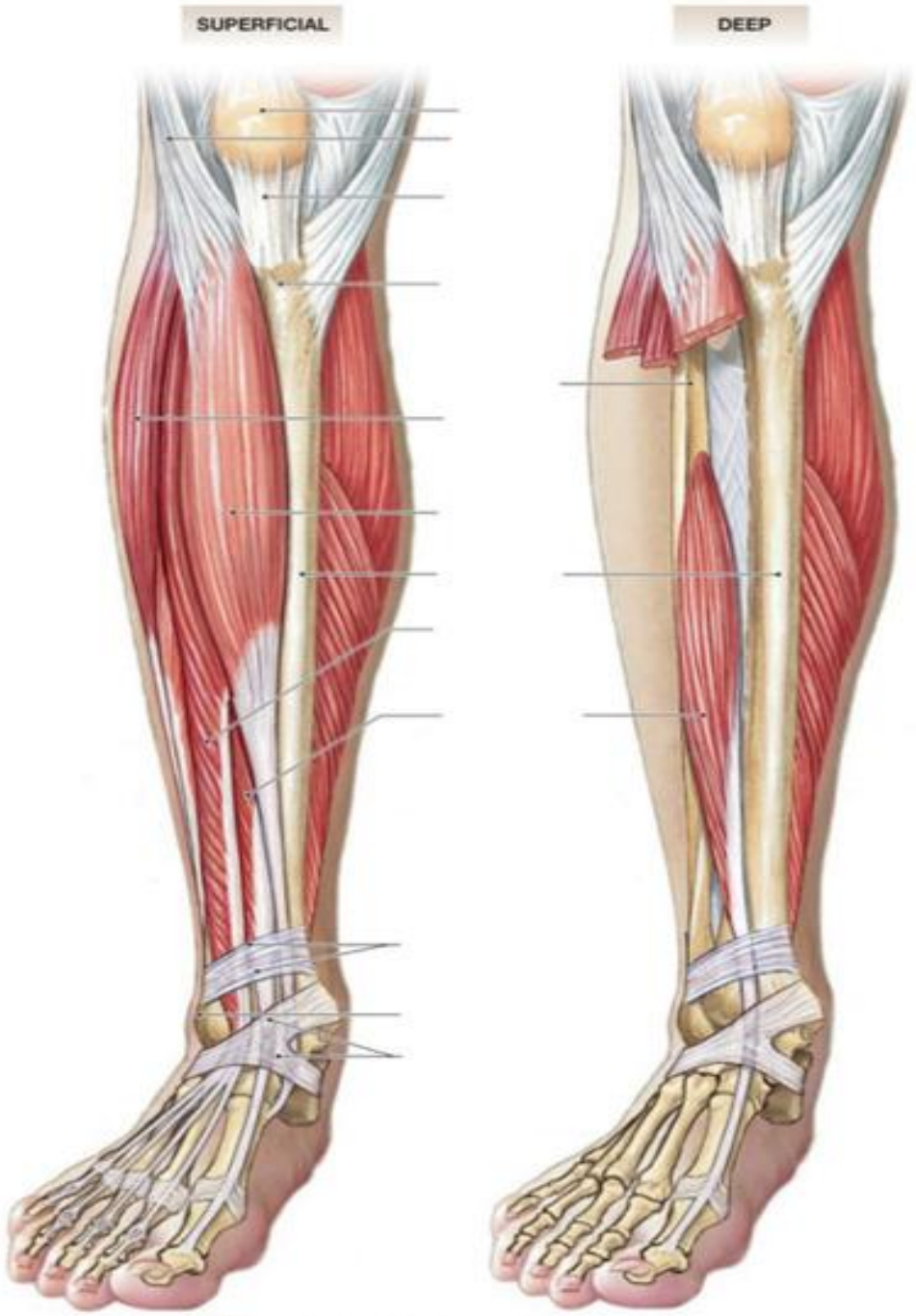


Assessment of the Ankle Joint

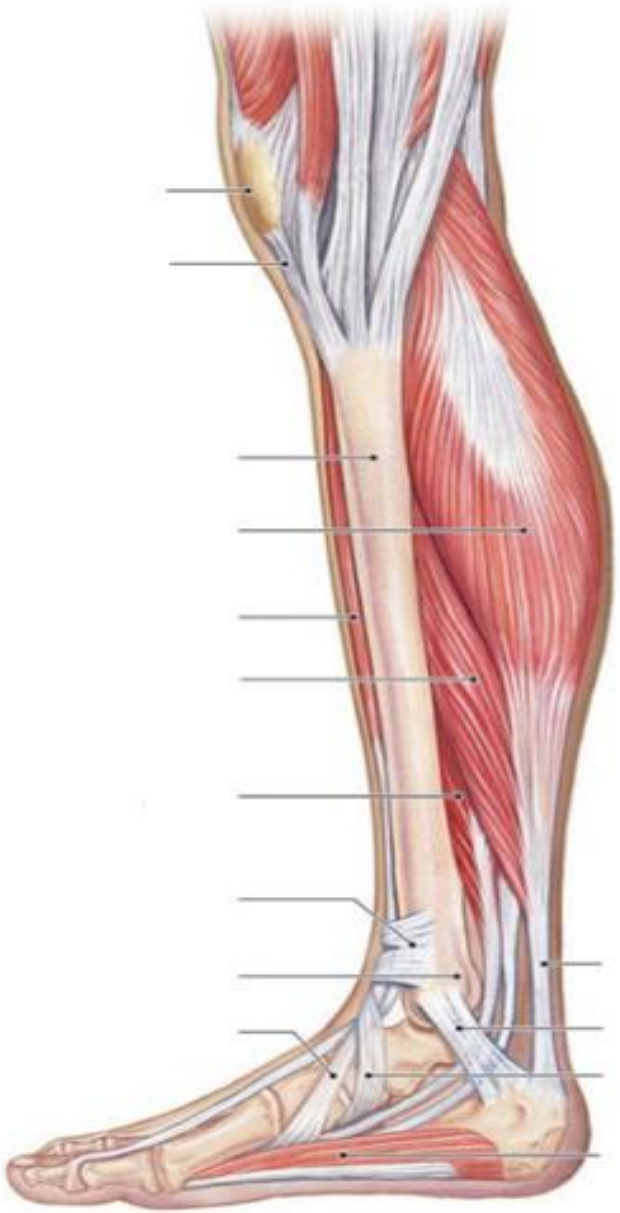
- joint Anatomy, including composition, bones, ligaments, Muscles, fascia, retinacluae ...etc
- Subjective assessment: Objective assessment:

Ankle joint muscles

- There are 3 Compartments of the muscles of the lower leg ankle joint:
    - Anterior compartment
    - Lateral compartment
    - Posterior compartment
- What are the muscles in each compartment? Review the origin and insertion of these muscles.
- Use the figure in the next slide to identify these muscles
- What are the main 2 ligaments on lateral and medial ankle joint?



a Anterior views showing superficial



a Medial view of the superficial muscles of the left leg



b Lateral view of the superficial muscles of the right leg

Subjective Assessment

Occupation? A lot of standing? Extensive Walking?	
Or Trauma (Direct or indirect)? Mechanism of Injury?	Is the joint swollen? Does the swelling occur with activity or several hours after activity, or does the joint feel tight at rest?
Has the ankle been injured before, or does it have any feeling of weakness?	Does the knee “give way?
Was there a “pop” when the injury occurred?	For active patients what type of workout they practice? Terrain or surface? Warm up and cool down?
Does the patient notice any deformity at the time of injury or after? Was the patient able to continue activity afterwards?	
Type of shoes they usually wear?	

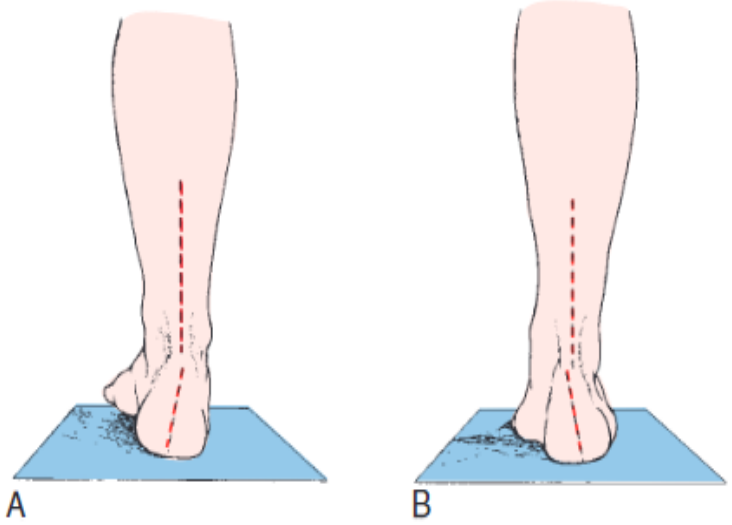
Objective Assessment

1- Observation

A. Informal: dynamic and static posture and movement, quality of movement…etc

B. Formal:

1. Observation of Posture and deformities.
2. Observation of muscle and skin condition.
3. Observation of soft tissue trauma.
4. Observation of balance and coordination.
5. Observation of gait and walking aids.
6. Signs of pain







- Supination or pronation of the foot (Pes Planus and Pes cavus)
- Achillis swelling
- Misshapen shoes due to severe foot pronation

Rom testing

What are the movements to assess?

1. Active ROM testing: using Goniometer (non wt bearing) and wt bearing.
2. Passive ROM testing : over-pressure and endfeel.
3. Accessory movement testing?

- Do you need to assess it?
- How will we use it?
- What are you looking for?

Practical: Assess ankle and subtalar joint ROM.



Active and Passive ROM

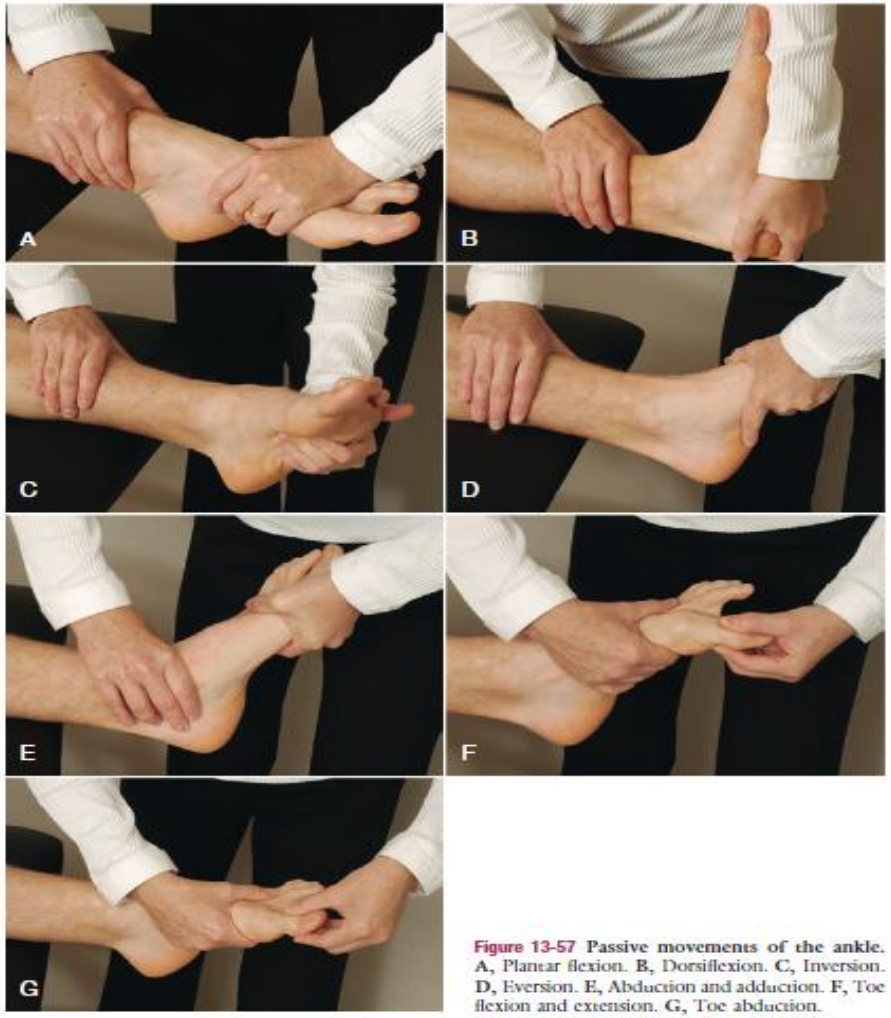
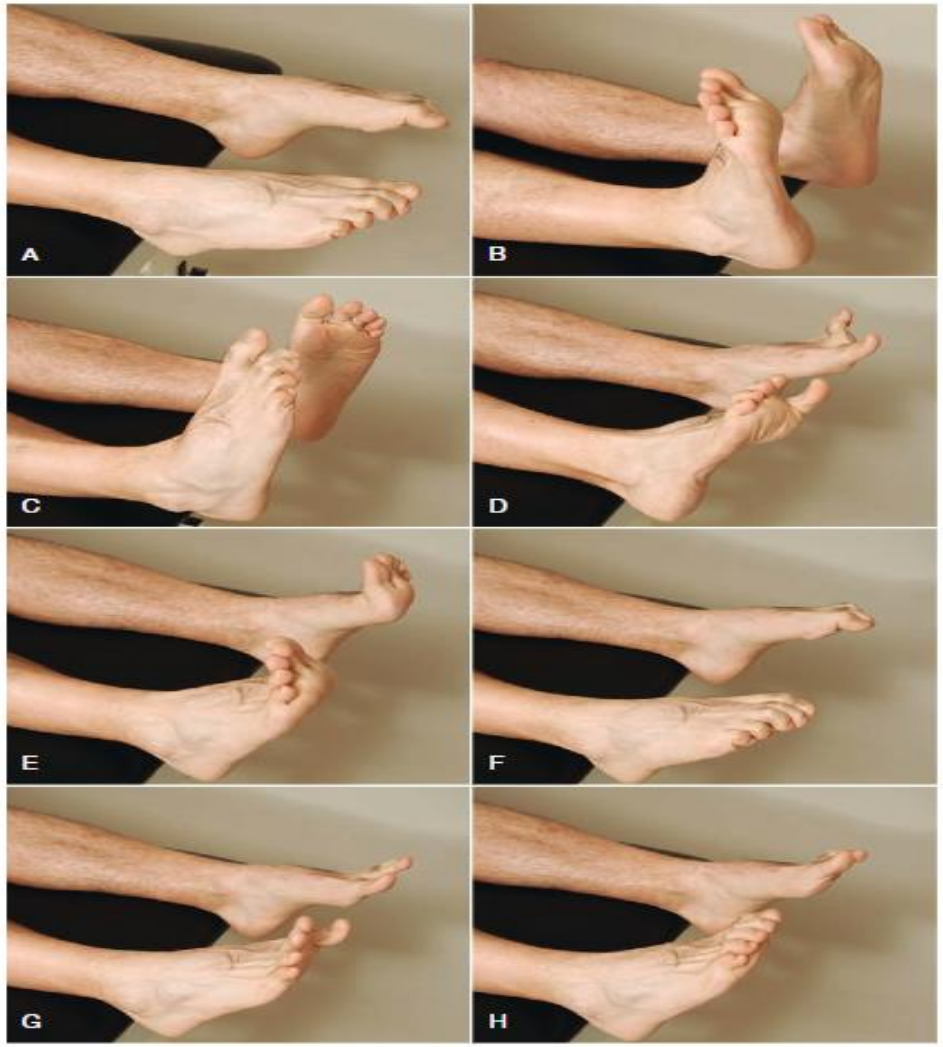


Figure 13-57 Passive movements of the ankle. A, Plantar flexion. B, Dorsiflexion. C, Inversion. D, Eversion. E, Abduction and adduction. F, Toe flexion and extension. G, Toe abduction.



Active ROM Ax

Movement	Patient	AXIS LOCATION	STATIONARY ARM	MOVEMENT ARM	ROM
Dorsiflexion	High- sitting	Lateral calcaneus at bisection of fibula and 5th metatarsal or over lateral malleolus	Parallel to fibula	Parallel with 5th metatarsal	20 degrees
Plantarflexion	High- sitting	Lateral calcaneus at bisection of fibula and 5th metatarsal or over lateral malleolus	Parallel to fibular	Parallel with 5th metatarsal	50 degrees
Inversion	Prone	proximal to Achilles tendon insertion	midline of lower leg	midline of calcaneus	20 degrees
Eversion	Prone	proximal to Achilles tendon insertion	midline of lower leg	midline of calcaneus	10 Degrees

Muscle Tests

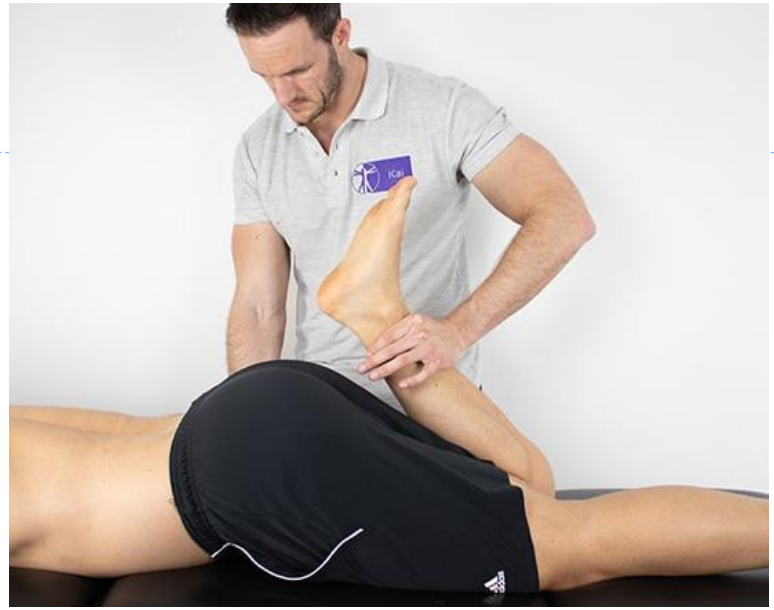
Which muscles to test their strength?

- Knee extension.
- Knee flexion (A).
- **Ankle dorsiflexion (B).**
- **Ankle plantar flexion (C).**
- **Ankle inversion (D)**
- **Ankle Eversion (E)**
- Big toe extension (F)



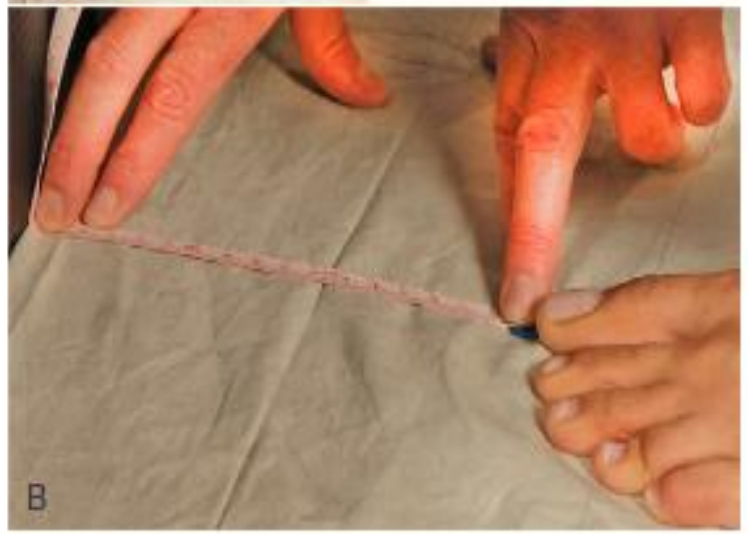
Muscle Tests

- 2- **Muscle length (flexibility):** What muscles to test or suspect their shortening.
- Ely’s Test
- Backsaver Sit and Reach
- **Ankle Lunge test**

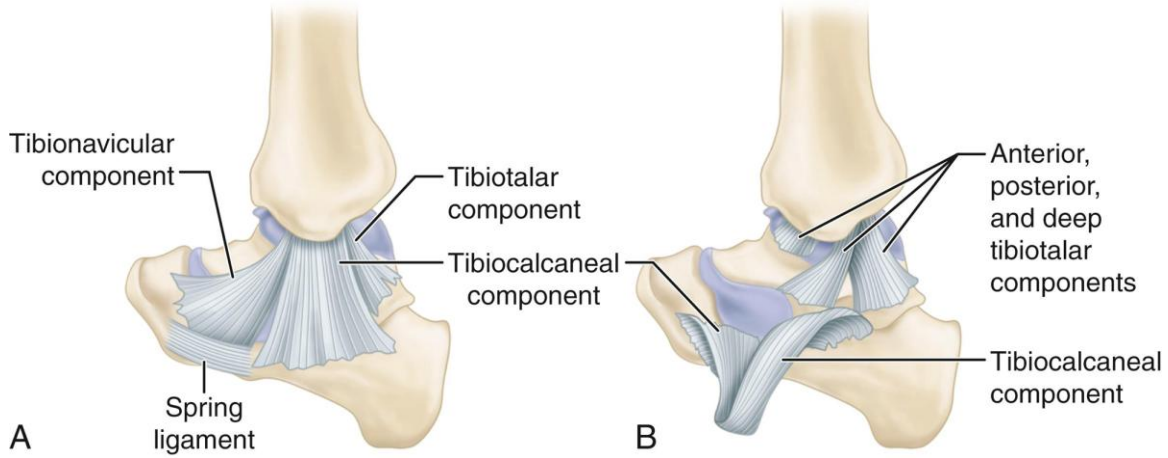


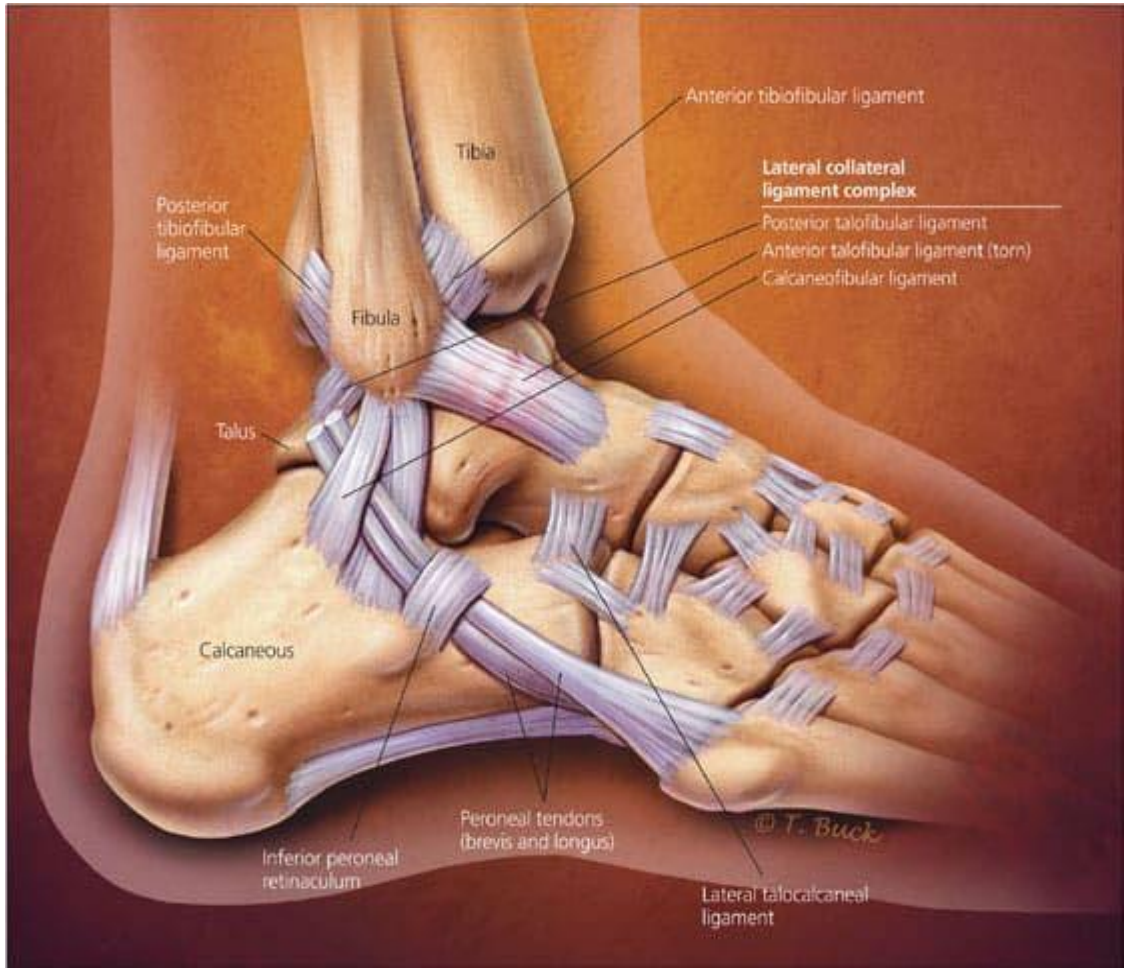






Lateral and Medial ankle ligaments



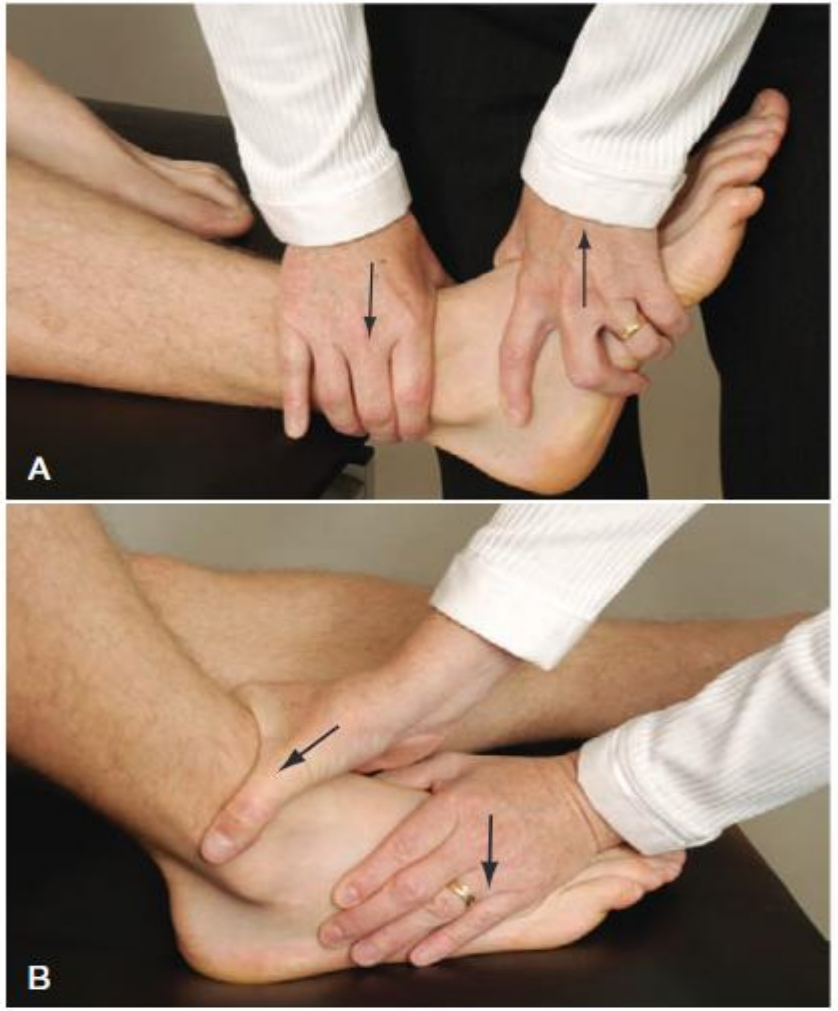


Anterior Drawer Test

A. Ligaments Integrity: Supine and prone Anterior TFL and acute joint



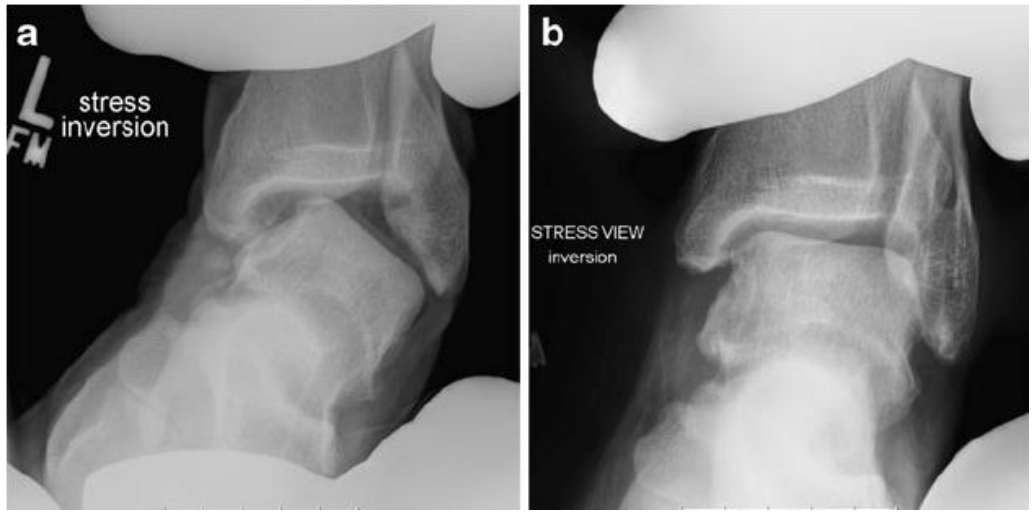
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Talar tilt test

Ligaments Integrity: Deltoid Ligaments and Anterior and posterior TFL and CFL



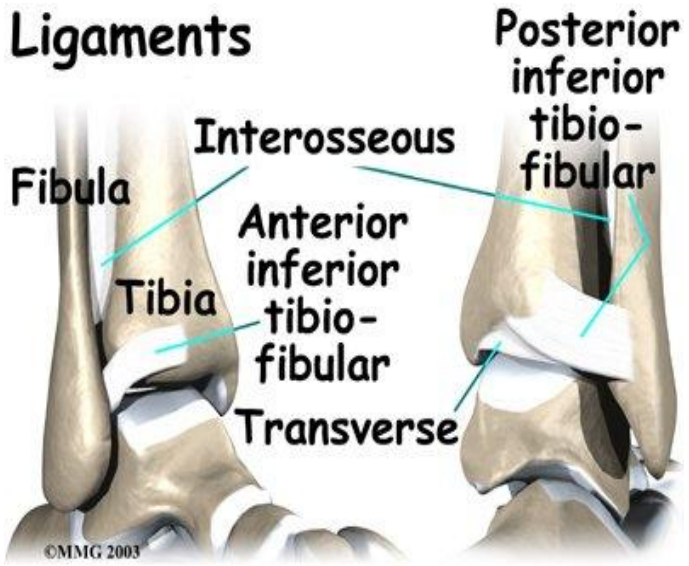




Assessment of syndesmosis

Syndesmosis Squeeze Test





Fibular translation test

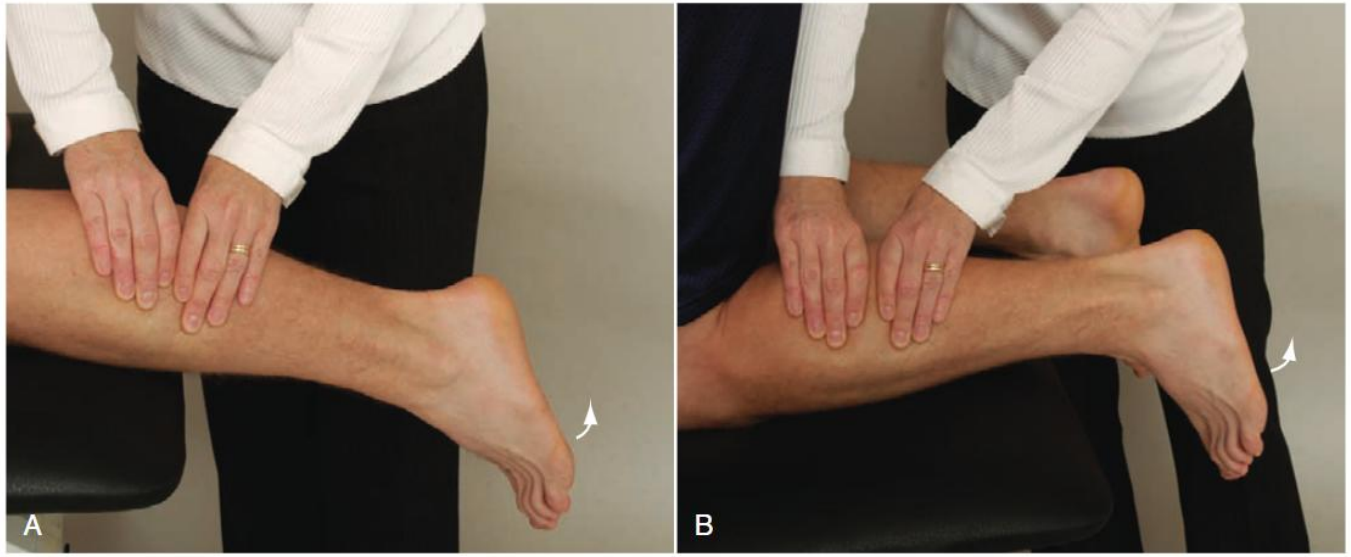


Achillis Tendon Rupture tests

Matles Test



Thompson test



Achillis Tendon Tear

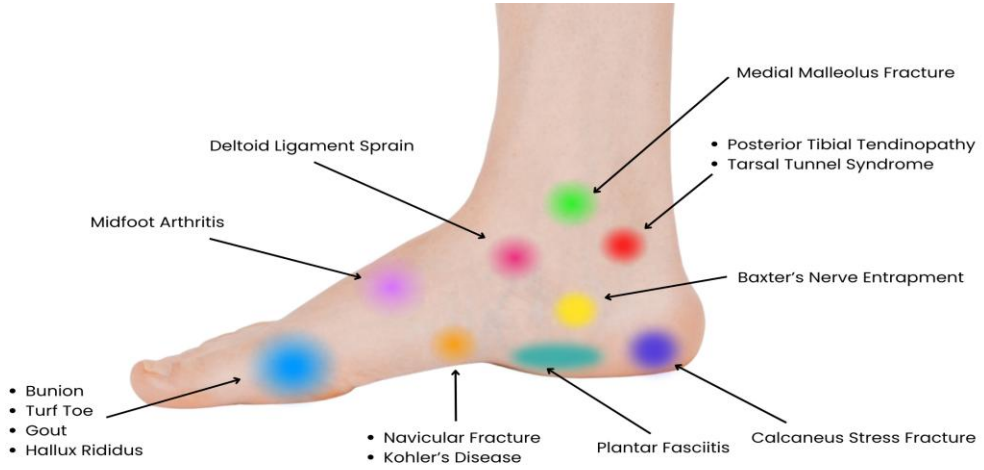


Windlass Test for plantar Fasciosis



Palpation

- Bony Prominence- Ottawa Ankle rule.
- Shin of tibia stress #
- Superficial dorsiflexors
- Superficial Planter flexors
- Ankle Vascularity



PALPATION OF LEG, ANKLE & FOOT

Palpating the Medial Malleolus

Positioning: client supine.

1. Locate the medial ankle with your fingertips.
2. Explore the large protrusion at the distal end of the tibia, called the medial malleolus.



Palpating the Calcaneus

Positioning: client supine.

1. Locate the posterior heel with your thumb and fingertips.
2. Pincer your thumb and fingers together, palpating the large, round calcaneus.



Palpating the Medial Tubercle of the Talus

Positioning: client supine.

1. Locate the medial calcaneus with your thumb.
2. Slide your thumb anteriorly and slightly proximally toward the medial malleolus onto the small, rounded medial tubercle of the talus.



Palpating the Sustentaculum Tali

Positioning: client supine.

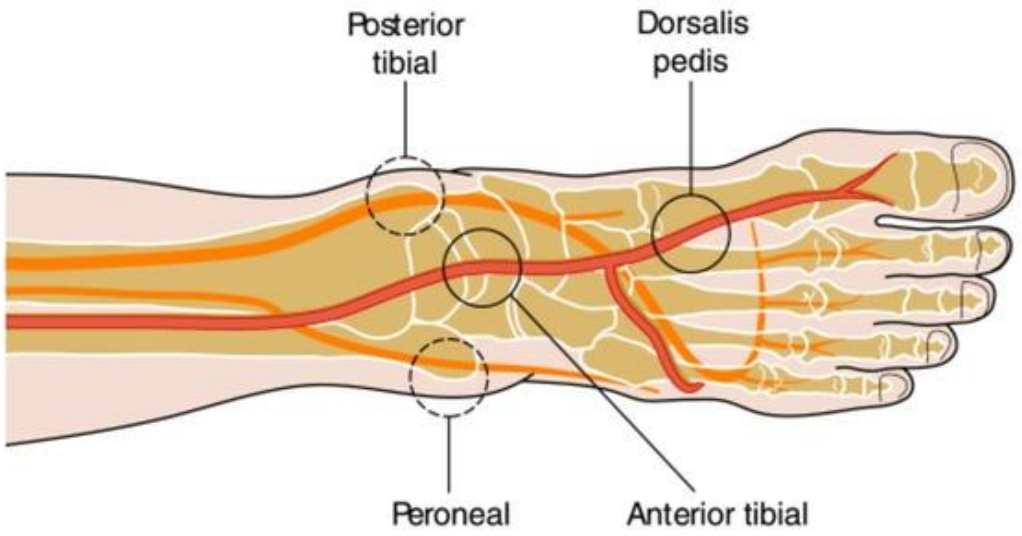
1. Locate the medial malleolus of the tibia with your thumb.
2. Slide your thumb distally, past the medial tubercle of the talus and onto the deep, blunt protrusion of the sustentaculum tali.



Stress fracture palpation



Testing Vascularity of the Ankle



Palpation of the Posterior Tibial Artery



Palpation of the Dorsalis Pedis Artery

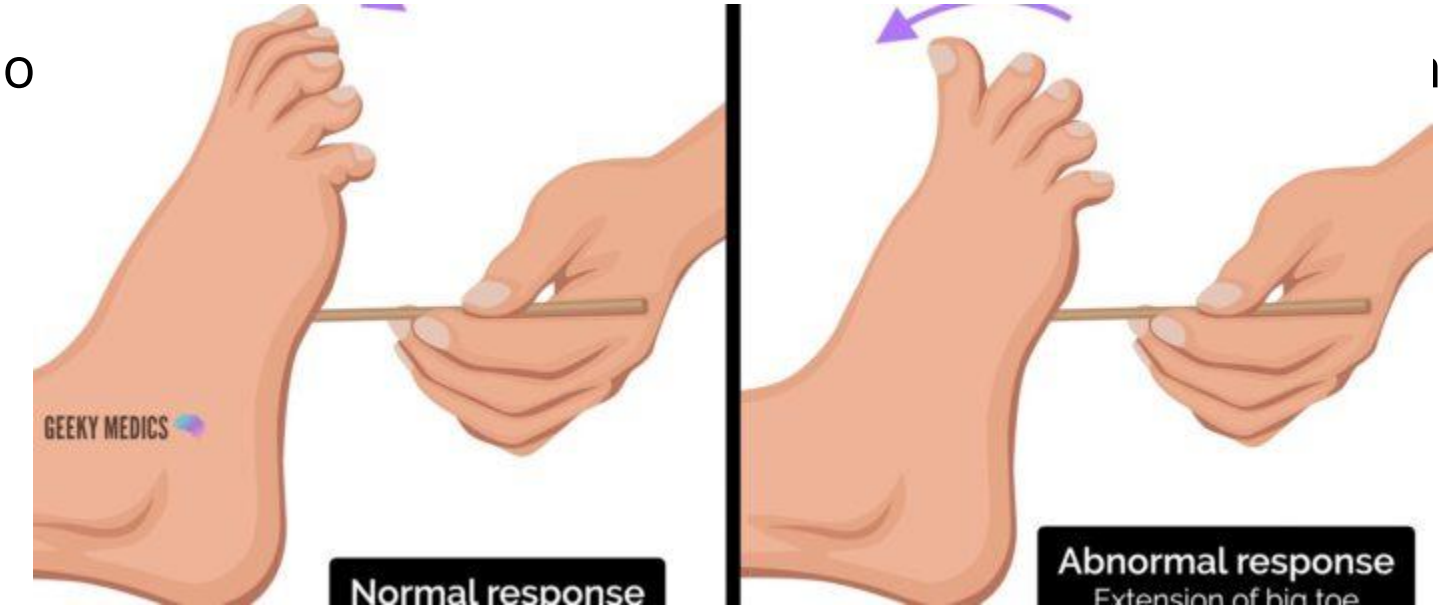


Figure of 8 test (Swelling)

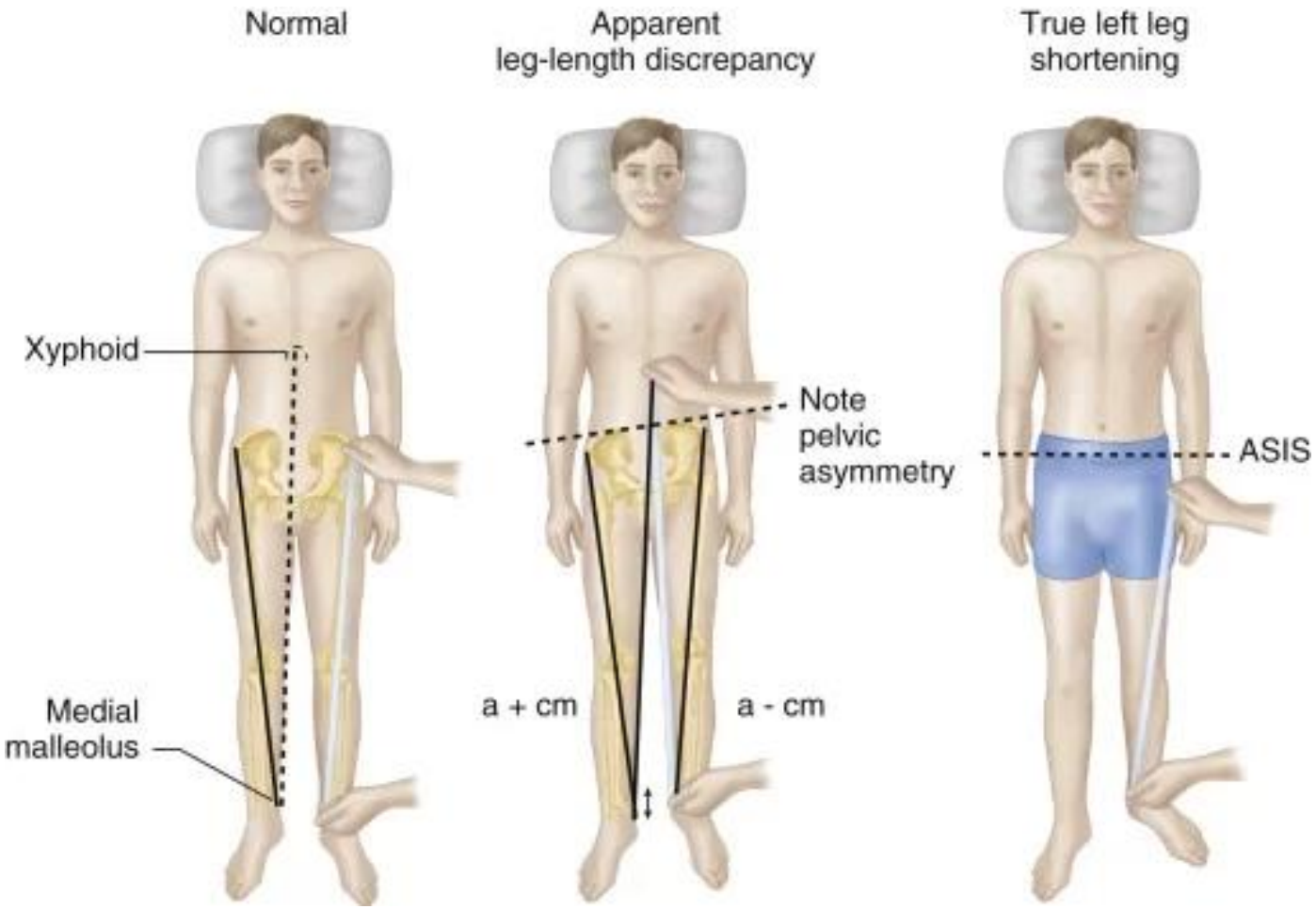


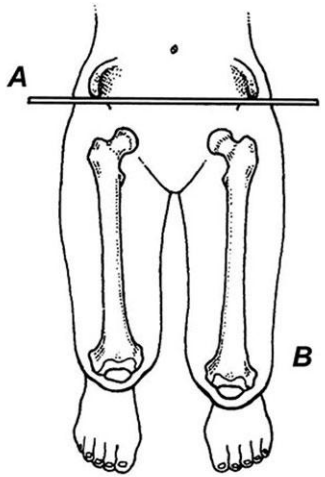
Neurological testing

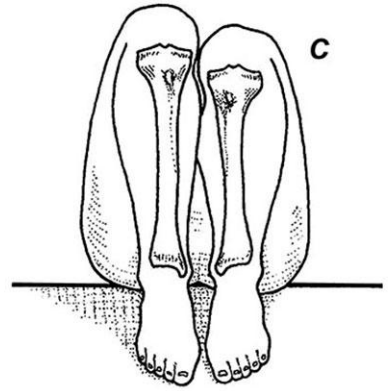
- Integrity of the nervous system: Dermatomes (Lumbar plexus) Myotomes (lumbar plexus) and reflexes (knee and ankle jerks) UMN screening (Babinski test, clonus)
- Dynamics o ıtic nerve)



Leg length discrepancy test







Functional testing

- Squatting post ankles
- Standing on toes
- Squat and pouncing at the end of the squat.
- standing on one foot, each at a time
- standing on the toes, one foot at the time
- going up and down stairs
- Walking on toes
- running straight ahead
- Running, twisting and cutting
- Jumping
- Jumping and going into a full squat

West Point Ankle Grading System (Ankle Sprain)

Criteria	Grade 1	Grade 2	Grade 3
Location of tenderness	ATFL	ATFL, CFL	ATFL, CFL, PTFL
Swelling & Ecchymosis	Slight, localized	Moderate, localized	Significant, diffuse
Weight bearing ability	Full weight bearing, Partial WB	Difficult without crutches	Impossible without pain
Ligament damage	Stretched	Partially torn	Completely torn
Instability	None	None - slight	Definite



Questions

